

Song–Chen Conjecture 2 at $n = 4$: a proof for uniform weights

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Abstract

Song and Chen refute the strong spin-alignment conjecture and leave open a weaker compatible-marginal majorization inequality, proving it at $n = 3$ for the maximally mixed reference. We settle the uniform-weight case at $n = 4$ with a complete argument, and reduce the general-weight case to a finite linear program that remains open. The enabling step replaces a grid argument — unsound here, because the slack is exactly zero at a vertex — with an exact rational enumeration over an arrangement with only seventeen box-vertices.

1. The statement

Conjecture 2 of [1] is a majorization inequality between a Hamiltonian assembled from compatible marginals and a reference spectrum. Song and Chen prove it for $n = 3$, maximally mixed reference, weights supported on 2-subsets [1, Prop. 3]. We take $n = 4$ at the same reference.

2. Uniform weights

The obstruction was a single tail inequality previously argued on a grid. Because the slack is exactly zero — attained at the all-zeros vertex — a grid cannot certify it, and the step has to be made finite and exact.

Write the relevant spectrum as $d(x) = 1 + \langle \varepsilon(x), u \rangle$ with $\varepsilon \in \{\pm 1\}^4$ and $u_q = \frac{1}{2} - r_q \in [0, \frac{1}{2}]$. The complementary pairing

$$d(x) + d(\bar{x}) = 2$$

holds exactly, so the sixteen values sort into eight pair-minima $m_p = (1 - a_p)_+$ lying below eight pair-maxima $M_p = 1 + a_p$, where $a_p = |\langle \varepsilon_p, u \rangle|$ ranges over the eight sign representatives. Each tail sum $\text{TailPlus}(r) = \sum_{i > r} \max(d_i, 0)$ then becomes a *minimum over which pair to drop*, hence piecewise linear on a sign-clip-box arrangement with exactly seventeen vertices — small enough to enumerate in exact rationals. All six tail inequalities hold, each meeting its threshold with slack exactly zero.

Clipping is essential: the naive bound $\sum_p m_p \geq 8 - \sum_p a_p$ fails, since $\sum_p a_p$ can reach 6. The vertex enumeration is therefore necessary rather than a convenience.

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3. General weights

The linear-programming route of [2, Thm. 3.2] applies and its validity was confirmed numerically to machine precision. The reduction to the target holds only after taking a minimum over the three matching-groupings: a fixed grouping overshoots the target. The remaining analytic obligation is therefore to show that for every weight and every state at least one of the three groupings stays under. This is a verified computational lead, not a proof.

4. Verification

The counterexample search ran on an independently constructed embedding agreeing with the original to machine zero, and the detector was validated by firing correctly on the paper's own $n = 3$ counterexample (top-three eigenvalue sum $0.8344\dots > 5/6$). Roughly 120k structured non-uniform weight configurations, a differential-evolution attack on the joint problem, and 60-digit arithmetic on the delicate cases returned no violation.

Disclosure

The mathematical content of this note — the construction, the case analysis, and the verification scripts — was produced by AI agents. My contribution was the choice of problem, the intuition for where a proof might come from, and the design of the adversarial procedure under which the claim was attacked and checked; I did not intervene in the derivation itself. Every load-bearing step was re-derived independently, in exact arithmetic where the problem allows it. This note is a record of a computer-assisted result. It has not been refereed.

References

- [1] Z. Song and L. Chen, *A counterexample to the strong spin alignment conjecture*, arXiv:2603.25410.
- [2] M. A. Alhejji and E. Knill, *Towards the spin-alignment conjecture*, arXiv:2307.06894.